

■■ SECURITY, AUDIT & LEGAL COMPLIANCE — TRACE FINDERS

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Compliance Target: Indian Evidence Act / Section 65B, IT Act 2000, Bharatiya Sakshya Adhiniyam 2023

Security Classification: Law Enforcement Forensic Operating Standard

■ Security Architecture Summary

- SHA-256 Chain of Custody Verification: Every evidence item ingested (audio recordings, CCTV video streams, bank statements, CDR CSVs) is assigned a SHA-256 cryptographic hash upon acquisition.
- Immutable Audit Trail: All workstation actions (viewing profile, context switching, graph exports, dossier generation) generate immutable audit log entries.
- Role-Based Access Control (RBAC): Enforces Analyst, Lead Investigator, and Judicial Reviewer permission levels.
- Data Isolation Policy: Person-scoped memory models prevent cross-case or cross-subject data contamination.

■ Cryptographic Hashing Protocol

```
interface AuditRecord {
  timestamp: string;      // e.g. "18 Aug 2026 09:43 IST"
  user_id: string;        // e.g. "INV-7092"
  action: string;         // e.g. "Switched Person Context"
  object_id: string;      // e.g. "P-002 (Rohan Mehta)"
  sha256_hash: string;    // e.g. "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
  result: "SUCCESS" | "DENIED";
}
```

■■ Legal Admissibility Compliance Matrix

Judicial Requirement	Technical Implementation in TRACE FINDERS	Status
Section 65B Certificate Support	Electronic record metadata, device node ID (#INV-7092), timestamp, SHA-256 hash	■ Compliant
Non-Tampering Proof	Read-only evidence drawer with hash verification status badge	■ Compliant
Audit Trail Immutable Logging	Operational Audit Explorer module tracking all analyst interactions	■ Compliant

